

EXPERIMENT - 9

TO DETECT ACID RADICAL PRESENT IN GIVEN SALT SAMPLE BY DRY AND WET WAYS. [SAMPLE A1]

APPARATUS REQUIRED

- | | |
|-----------------------------|-----------------------------|
| a) <u>Test Tube</u> | b) <u>Test tube holders</u> |
| c) <u>Burner</u> | d) <u>Test tube stand</u> |
| e) <u>Test tube brushes</u> | f) <u>Wash bottles</u> |

CHEMICAL REQUIRED

- | | |
|-------------------------------|---------------------------------------|
| i. <u>dil. HCl</u> | ii. <u>Conc. H_2SO_4</u> |
| iii. <u>Copper burnings</u> | iv. <u>dil. HNO_3</u> |
| v. <u>$BaCl_2$</u> | vi. <u>$FeSO_4$</u> |

THEORY

qualitative salt analysis. It is the detection of the acidic and basic radicals present in given salt sample.

Quantative salt analysis: It is a branch of chemistry that deals with the study of measurement of qualitatives of particular constituents present in the substance.

Acid - Acid are those substances which react with bases to give salt and water and ionise to give H^+ ions.

Base - Base are those substance which react with acid to give salt and water and ionise to give OH^- ion.

Acidic radicals - The electronegative radicals are called acidic radical. e.g. H^+ , Cl^- , F^- , O^- etc.

Basic radicals - The electropositive radicals are basic radicals. e.g. Na^+ , Ca^{++} etc.

Dry test - The test which are performed directly with sample in dry and solid state.

Wet test - The test which are performed with the given sample in the form of solution with some suitable solvent.

SAMPLE NO: A₁

State: Crystalline

Colour: Colourless

Odour: Odourless

Solubility: Soluble

OBSERVATION TABLE (Dry test for acid radical)

S.N.	Experiments	Observations	Inferences
①	Dilute mineral acid test A pinch of given salt was kept in a clean dry test tube and few drops of 'dil. HCl' was added	No gas colour was seen	absence of NO_2^- , S^{--} , SO_3^{--} , CO_3^{--}
②	Conc. H_2SO_4 test A pinch of given salt	No any gas was evolved	Absence of Cl^- , Br^- , I^-

S.N.	Experiments	Observations	Inferences
	was given added into a clean dry test tube & few drops of H_2SO_4 was added		
③	Dry test for nitrate Little sample of salt in clean and dry test tube was added & few drops of conc. H_2SO_4 along with some pieces of Cu turning was added and heated	Brown fumes evolved that turned ferrous sulphate solution to black	Presence of NO_3^-

WET TEST FOR ACID RADICAL

A pinch of given salt sample was taken in a clean test tube and was dissolved in minimum quantity of suitable solvent & diluted by adding more solvent in it.

This original solution (O.S) of salt was used to perform the test.

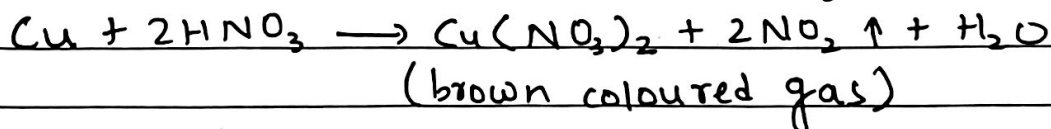
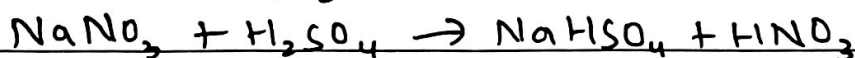
OBSERVATION TABLE (Wet test for acid radical)

S.N.	Experiments	Observations	Inferences
	1 ml of Original soln was taken in a clean test tube & 2 ml of conc. H_2SO_4 was added. The content content was cooled under tap water. Freshly prepared $FeSO_4$ soln was added through the wall of test tube without disturbing the mixture solution.	A dark brown ring appeared at the junction of the two liquid layers.	Presence of NO_3^-

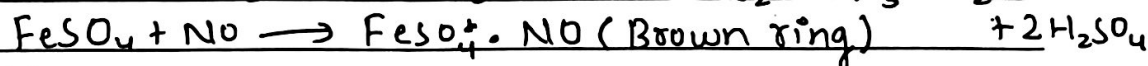
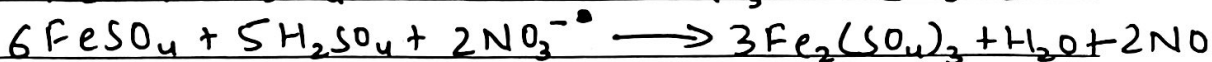
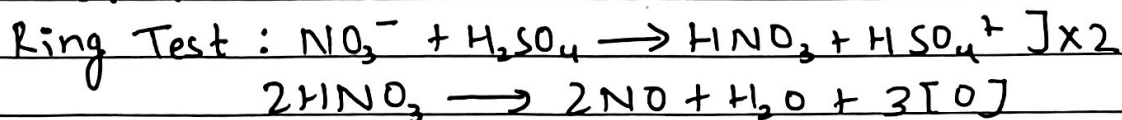
TEST REACTIONS

A. Dry test

- Conc. H_2SO_4 reacts with nitrate salt to give HNO_3 which react with Cu to give brown coloured NO_2 gas



B. Wet test



RESULT

In this way, ~~acidic or basic~~ acidic or basic or complete salt is detected to be nitrate (NO_3^-)

CONCLUSION

Hence, the above experiment can be concluded that acidic or basic radical or complete salt in the given sample can be detected both in dry and wet ways.

PRECAUTIONS

- ① Personal protective equipment should be worn.
- ② Apparatus should be handled with great care.
- ③ Reagents and chemical should be used minimum quantity as far as possible.

VIVA-VOCE

1. What is qualitative salt analysis ?
2. What are acid and basic radicals?
3. What is dry test and wet test?
4. What happens when the AgCl ppt. obtained by treating O.S. with AgNO₃ is reacted with excess of ammonia solution?

① It is the detection of the acid and basic radicals present in given salt sample.

② Electro-negative species of salt derive from acid is called acid radical.

E.g: Cl^- , SO_4^{2-} , NO_3^- etc.

Electro-positive species of salt derive from base is called basic radical.

E.g: Na^+ , NH_4^+ , Al^{+++} , Mg^{++} etc.

③ The test performed directly with sample in dry and solid condition is called dry test. While the tests which are performed by involving water is called wet test.

